

QUOC ANH TRINH (ANDREW)

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FINANCIAL CRIME SOLUTIONS

Financial crime solutions specialist who builds – not just operates – **Machine Learning**, **AI**, and **LLM**-powered systems. Engineers the optimum solution at minimum cost: free-tier and open-source tooling first, paid infrastructure only once value is proven. Deep in **Agentic AI**, **LLM** tooling, and **BigQuery ML**, with **SQL**, **Python**, **R**, **Power BI**, **Tableau**, **AWS**, **GCP**, **Oracle SQL**, and **NetReveal/RSA-Outseer** across banking, FinTech/crypto, and investment banking (**Bendigo & Adelaide Bank**, **Remitano**, **National Citizen Bank**). Proof in the build: self-directed AI products **Typology Extractor** and **PheChat**, shipped lean.

KEY SKILLS

AI & Machine Learning: (Python, LLMs, NLP, Agentic AI, BigQuery ML)	• Built Typology Extractor (Project) — an Agentic AI/ETL pipeline transforming AUSTRAC guidance into structured data, supporting enforcement-gap detection & national scenario benchmarking • Co-delivered UAR-to-SMR AI Automation (Pydantic-validated LLM prompting), cutting reporting time 80%; built PheChat (Project), an Agentic RAG chatbot • Certified: Claude Code, Claude API, Agent Skills & Subagents (Anthropic) and BigQuery ML Skill Badges (Google) (Certificates)
Statistics & Detection Systems: (SQL, Python, R, statistical calibration)	• Scenario Management: build, review & calibrate detection scenarios via optimisation algorithms & statistical analysis • Scenario Development Coordination (2025): designed & calibrated 9 new detection scenarios, personally delivering 3 with a statistically significant true positive uplift • Annual Rule Review: statistical testing & sampling cut false positives 5%, removing 5,000+ alerts
Automation: (Oracle SQL, ETL, RPA-style pipelines)	• Manual Alert Ingestion Pipeline: Oracle SQL automation feeding 150+ alerts daily into NetReveal & RSA/Outseer with 99.9% data integrity • UAR-to-SMR Automation: data masking/rehydration, cutting reporting prep time 80% • Automated email marketing (30% CTR uplift); optimised ETL, cutting workload 25%, cloud costs 30%, runtime 70%
Dashboard Building: (Power BI, Metabase)	• Dashboard & Reporting: Power BI dashboards & monthly reporting presented at Board level, eliminating 70% of ad-hoc requests and cutting prioritisation time 60% • Significant Dashboard Portfolio: IFTI SWIFT network graph & BEN outlier detection dashboards spanning operational & detection uplift • Infrastructure Establishment: stood up the team's Power BI Server & cloud workspace
Leadership: (Technology, Operations, Data, Group Risk)	• Scenario Development Coordination (2025): personally delivered 3 of 9 scenarios and supporting others • Supported the Annual Rule Review & led the GitLab Migration, improving version control & governance

PROJECTS

- **Typology Extractor** (typologyextractor.vercel.app) — Agentic AI and ETL pipeline that transforms AUSTRAC guidance documents into structured data for transaction monitoring framework comparison. Supports enforcement-gap detection and national scenario benchmarking across financial crime typologies.
- **PheChat** (phechat.com) — Agentic RAG chatbot built for small and medium businesses, combining retrieval-

augmented generation with autonomous agent workflows to answer customer queries and automate routine business tasks.

CERTIFICATES

- **AI Introduction to Subagents** – Anthropic, Jul 2026
- **AI Building with the Claude API** – Anthropic, Jul 2026
- **AI Introduction to Agent Skills** – Anthropic, Jul 2026
- **AI Claude Code in Action** – Anthropic, Jul 2026
- **AI AI Fluency Framework & Foundations** – Anthropic, Jul 2026
-  **Engineer Data for Predictive Modeling with BigQuery ML** – Google Skill Badge, Jun 2025
-  **Create ML Models with BigQuery ML** – Google Skill Badge, Jun 2025
-  **Modernizing Data Lakes and Data Warehouses with GCP** – Coursera, Jun 2020

PROFESSIONAL EXPERIENCE

Analyst **Bendigo & Adelaide Bank** **05/2021 - Present**

Intelligence and Analytics - Financial Crime Risk

Big 5 Australian banking group

- **Scenario Management:** Build, review, and calibrate transaction monitoring detection scenarios, applying re-search and statistical analysis to drive the evergreen assessment of screening and detection systems.
- **Solution Delivery:** Design and deliver financial crime detection solutions—including the AI-driven UAR-to-SMR automation pipeline and manual fraud/AML detection scenarios built outside vendor platforms—partnering across Technology, Operations, and Data teams.
- **Framework Set-Up:** Set up and maintain the Power BI workspace and GitLab space underpinning the transaction monitoring product.
- **Dashboard Building:** Own the Power BI dashboards as part of the transaction monitoring product—designing, developing, and maintaining them to provide key measures, KPIs, and guardrails.

ACHIEVEMENTS

- **Annual Rule Review:** Executed and peer-reviewed the annual transaction monitoring rule review in **2022** and **2023**, supporting team members with statistical confidence testing, data sampling, and SQL-based scenario recalibration to cut false positives by **5%**, removing **5,000+** unnecessary alerts and saving **300+** investigation hours a year.
- **Scenario Development Coordination (2025):** Executed and supported team members in building **9** new detection scenarios in **2025**—personally delivering **3** with a statistically significant true positive uplift—while providing support on datasource, design and calibration.
- **UAR-to-SMR Automation (2025):** Co-delivered an end-to-end **AI-driven UAR-to-SMR submission automation** pipeline—implementing secure **data masking/rehydration** and **Pydantic-validated LLM prompting**—cutting regulatory reporting preparation time by **80%** while ensuring strict transparency and data privacy compliance.
- **Manual Alert Ingestion Pipeline:** Developed, implemented, and maintained a manual fraud/AML alert ingestion pipeline for detection scenarios built outside vendor platforms, feeding **150+ alerts daily** into **NetReveal** and **RSA/Outseer** with **99.9% data integrity**.
- **Rapid Incident Response:** Stood up a full detection service from scratch within **2–3 days** to support an urgent brand transition and incident response, maintaining uninterrupted transaction monitoring coverage under significant time pressure.
- **GitLab Migration:** Migrated manual detection scenarios from a shared network drive to **GitLab**, improving version control, auditability, and scenario governance.
- **Dashboard & Reporting:** Built Power BI dashboards and monthly reporting used by Operations and presented at **Board level**, eliminating **70%** of ad-hoc requests and cutting prioritisation time by **60%**.
- **Significant Dashboard Portfolio:** Delivered a suite of high-impact Power BI dashboards spanning **operational uplift** (UAR, TM, and WLM case prioritisation) and **detection uplift**, including an **IFTI SWIFT network graph** for cross-border payment analysis and a **BEN outlier detection** dashboard for anomaly identification.

- **Infrastructure Establishment:** Initiated and established the team's **Power BI Server**, **Power BI cloud** workspace, and **GitLab** environment—foundational infrastructure enabling scenario version control and dashboard delivery.

Data Analyst

Remitano

03/2020 - 04/2021

International cryptocurrency exchange that matched \$500M+ in trading value and operated in more than 30 countries.

- Owned all data analytics requests, delivering dashboards and diagnostic reports in Python, SQL, and Metabase.
- Built marketing predictive models to automate and optimise strategy deployment.
- Designed, developed, and managed data crawlers, ETL pipelines, and Redshift database structures.
- Collaborated with engineering teams on complex SQL-based data transformations for scalable operational and marketing analytics.

ACHIEVEMENTS

- Automated email marketing via customer classification models (Python/SQL), boosting click-through rates by **30%** and enabling scalable campaign personalisation.
- Optimised ETL pipelines, indexing, and data normalisation in Redshift, slashing developer workload by **25%**, cloud costs by **30%**, and dashboard runtime by **70%**.
- Delivered **100% on-time** analytical reports and dashboards with consistent stakeholder satisfaction scores of **8/10 or higher**.
- Engineered complex SQL solutions (including recursive queries) for MLM commission calculations, processing **150K+** referral relationships across **4** levels to simplify payout computation, eliminate the need for a graph database, and power a **4-tier** referral program that drove **25%** of new user growth.

Investment Associate

National Citizen Bank

09/2014-02/2018

National Citizen Bank – A Commercial Bank with more than 20 years of operation

- Built financial models for property project estimation, company valuation, and debt restructuring, presenting findings to the Investment Council to support investment recommendations.
- Coordinated with in-house executives and clients on lending due diligence, gathering business information to support financial analysis and valuation.

ACHIEVEMENTS

- Secured Investment Council approval for **80%** of recommendations, generating a public equity portfolio **CAGR of 10%**.
- Cut investment approval time by **72 hours** and eliminated process conflicts by drafting a formal investment procedure covering accounting journal entries, authority levels, and document revision.

HONOURS & AWARDS

- Third place & Honourable Mention - Project SafeAus - Making Australia safer, Team VAIT - GovHack Australia, 09/2021
- #13 in public test and #32 in private test - PetFinder Competition, Team Vicohub-1 – Kaggle, 02/2019-03/2019
- First prize & Runner-up - Programming Competition – Department of Education and Training – 2007, 2008 & 2009

EDUCATION

- Master of Business Analytics – Deakin University, 2019
- Bachelor of General Business Management – National Economics University, 2013

VOLUNTEER ACTIVITIES & HOBBIES

- Tutored 100+ students in SQL and Python
- Won 2 Open Gov hackathon prizes
- Served as Coordinator of CFA Vietnam volunteer team
- Hobbies: Sports (Football/Soccer, Powerlifting), Building data-driven solutions for micro-businesses

REFERENCES

Available upon request